

2014

(i)

Low Temperature flame for alkali and alkaline earth metals

Alkali and Alkaline earth metals have low excitation energies. This is the fact that compounds of alkali and alkaline earth metals are thermally dissociated into atoms at the Temperature of Bunsen burner flame and that some of these atoms produced are further excited to higher energy level. When these excited atoms return to the ground state, they emit radiation which for these elements lies in Visible region of electromagnetic spectrum.

(ii)

Function of flame in Flame Photometry

- 1- To convert the analyte of the liquid sample into vapour state.
- 2- To decompose the analyte into atoms and simple molecules.

- 3- To excite the formed atoms/free atoms simple molecules to emit radiation energy.

(iii)

Electrophoretic mobility

The movement of ions under the influence of electric field. In this case, negative ions move toward anode and vice versa. It is represented by μ .

$$\mu = V/E$$

" the rate of migration (cm/sec) per unit field strength. (Volts/cm)

$$\mu = Q/6\pi r\eta$$

Q = net charge on ion

r = ionic radius

η = Viscosity of the medium.

Electroosmosis mobility (Electroosmotic flow)

It simply refers to the movement of liquid in a porous

material due to applied electric field.

In Electro-osmosis the solid which is the channel wall is stationary.

Mathematically,

$$u_{EOF} = -\frac{\epsilon \zeta_0 E}{\mu}$$

ζ_0 = zeta potential.

E = electric field.

μ = dynamic viscosity

ϵ = relative permittivity

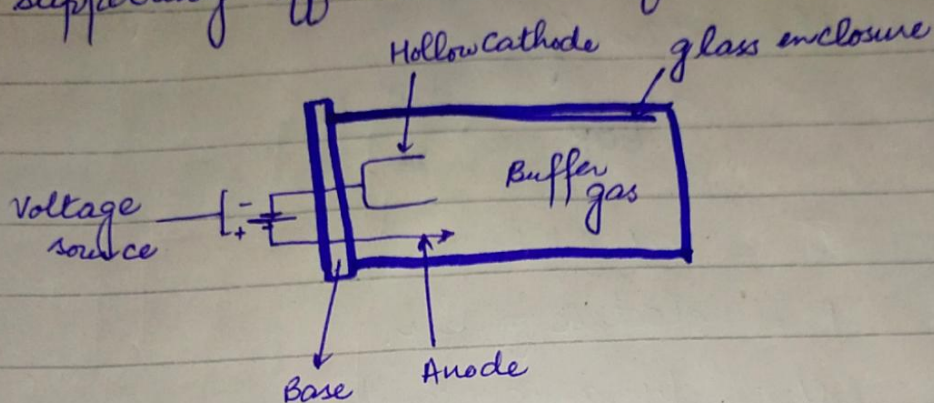
this phenomenon is useful in chemical separation techniques and buffered solution.

(iv)

AAS limited to metals

This limitation is due to Hollow cathode lamps used as a light source, these lamps are built by using metals you are analysing for and have electrically conductive to work as a cathode in lamp. As it is obvious that cathode made from the element of

interest and an anode also. Inert gas is used in tube when inert gas ions accelerate into cathode supporting off atoms occur from cathode.



(V) Limitations of Flame Photometry

- 1- The number of excited atoms in flame is very small. It is the alkaline and alkaline earth metals that can be practically determined.
- 2- It needs perfect control of flame temperature.
- 3- Interference by other elements is not easy to be eliminated.

4. Heavy and transition metals, the number of absorption and emission lines are enormous and the spectra are complex.

(vi)

Ion exchanger usually swells in H_2O .

As the ion exchange is a reversible interchange of charged particles or ions with those of like charge. This occurs when ions present on an insoluble resin matrix effectively swap places with ions of similar charge that are present in surrounding solution. Swelling of ion exchanger is primarily a hydration of **fixed ionic groups** and increases with an increase in capacity to the limits imposed by the polymer network.

(vii)

Packing of column with dry gel.

The surface of resin must be wet. If the resin dry up or we use dry gel channels or cracks may be

formed in the resin bed. In this case, there will be incomplete contact between the solution and resin when column is subsequently used. In preparing ion exchange column, the resin must be fully hydrated with deionized water because if it is put on the column in dry form, swelling pressure built up and cause cracking in resin column. or it is also possible the column will burst on adding H_2O .

(viii)

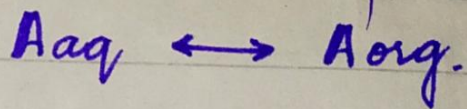
distribution ratio, at const Temp

As distribution ratio is a constant value, It is equal to the concentration of solute in organic phase divided by its conc. in the aqueous phase. It is based on the Nernst distribution Law which is applicable on constant Temperature. also depends upon distribution law. According to this Law, $C_1/C_2 = K$. And value of K depends upon Temperature, nature of solute, nature of two immiscible solvents.

(ix)
Distribution Law.

This Law states that
"When a solute distributes itself between two immiscible phases, the ratio of concentration in two phases is always constant, at constant Temperature."

There should be an equilibrium between two immiscible phases.



(X)
Ion Exchange Chromatography.

Ion Exchange Chromatography is a form of Solid-liquid chromatography in which the stationary phase is an ion exchange resin. Most of these resins consist of insoluble polystyrene organic polymers linked (cross-linked) with divinyl benzene. The ions present in the solution with similar charges act as mobile phase.

There are two types of Ion Exchanger

- * Cation Exchange Resin
- * Anion Exchange Resin

2015

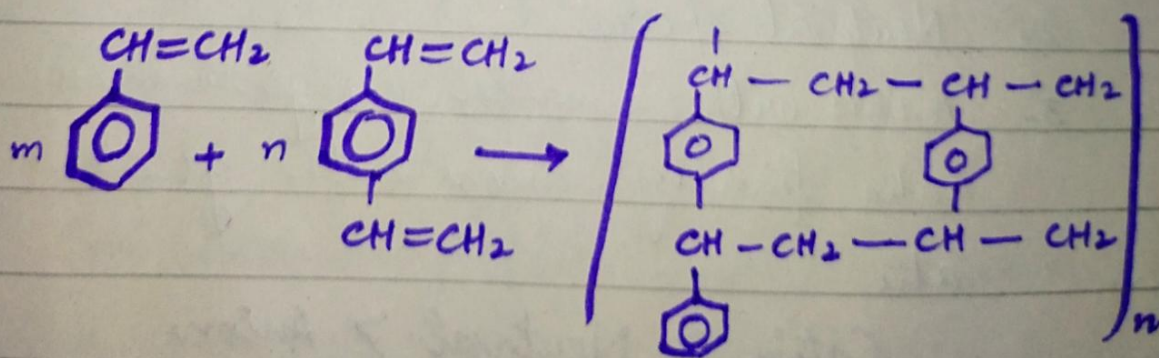
Short Questions

(i)

Cross-Linkage in Ion Exchange Resin.

In Ion Exchange chromatography, Ion exchange resin is used as stationary phase. Most of the Ion Exchange Resins are insoluble polystyrene organic polymers cross-linked with divinyl benzene with a large number of ionic functional attached covalently with free phenyl group present in it.

It is a long twisted chain of Carbon with benzene ring at every other carbon. It will form a bead-like three dimensional structure.



Q no 2, 3, 4 in 2014's paper.

Q (5)

Order of species separated in normal Capillary zone electrophoresis.

Ionic species can be separated on the basis of their charge, radius and frictional forces by capillary zone electrophoresis. In an applied electric field, the differential movement of analytes is the basic principle of CZE.

- 1- Cations with largest Charge-to-mass ratios separate out first, followed by Cations with reduced ratio
- 2- Neutral species
- 3- Anions with smaller charge to-mass ratio finally anion with greater ratio

Cation > Neutral > Anion

Q (6) Effect of Electrophoretic mobility on the separation in CZE

As the electrophoretic mobility is depend on charge of molecules, viscosity and atomic radius. A mixture in a solution can be separated into the individual components quickly and easily. This separation is based on the differences in electrophoretic mobility which is directly proportional to charge on the molecule and inversely proportional to the viscosity of the solvent and radius of atom.

$$\mu = Q / 6\pi r \eta$$

Q no 7, 8 in 2014's paper

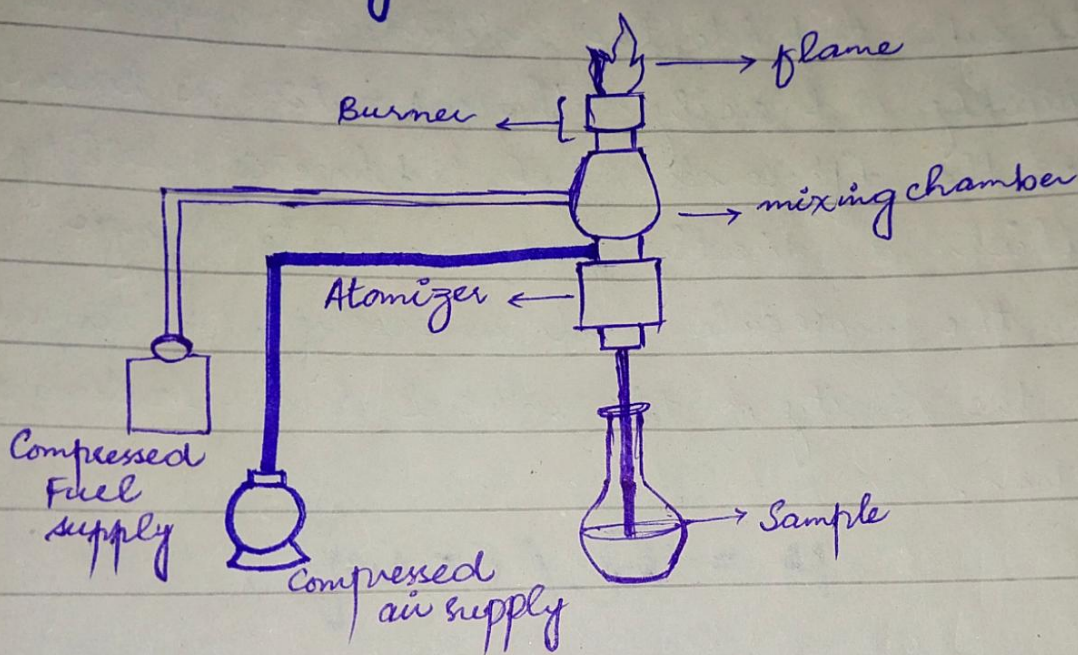
Q 9 Gel Chromatography as size exclusion

It is a type partition chromatography in which the separation of the compounds is obtained on the basis of molecular size and shape. Separation is achieved by

differential exclusion from the pores of the packing material, of the sample molecules as they pass through a bed of porous particles

Q 10

Burner system in Flame Photometry



2017

Short Question

(i)

Synergistic Extraction

This extraction has become a common method for separation of metal ions. It can't only improve the extraction efficiency but also enhance the stability of extracted complexes in organic phase.

Synergistic agents are added to organic phase in order to increase the extraction. They form complexes which are taken by the extractant, even if the extractant does not form suitable complexes directly with the desired metallic species. Neutral organophosphorous compounds are added to acidic organophosphorous compounds to exert synergistic effect.
example.

Synergistic effect is the combination of two or more drugs effect that causes the total effect of drugs to be greater than the sum of individual effects.

(ii) Electroosmotic flow

"It is defined as the movement of liquid through porous substance under the influence of electric field"

In case of CZE, capillary tube membrane or other liquid is used.

When high voltage is applied across a capillary tube made up of fused silica containing buffer solution, excessive cations are induced to migrate towards Cathode. This migration in capillary tube is called Electro-osmosis (flow) & flow of bulk solvent is called electro-osmotic flow.

$$\mu_{EOF} = \frac{\epsilon \epsilon_0 E Z}{4 \pi \eta}$$

(iii) Characteristics of Cation Exchange resins.

Types of Cation Exchange Resin

① Strong Acid

② Weak acid

For Strong Acid Cation exchanger

- ① → Highly ionized in both acid and salts from the sulphonic group
- ② → Exchange capacity of strong acid resins is independent of the solution pH.
- ③ → Used in the hydrogen form for complete deionization.
- ④ → Used in Sodium form for Softening of water (Ca^{2+} & Mg^{2+} removal)

For Weak Acid Cation exchanger

- ① → Ionizable group is a carboxylic acid
- ② → The degree of dissociation is strongly influenced by the solution pH.
- ③ → limited capacity below a pH of 6.0

(iv)

Multiple-Extraction in Solvent extraction.

When the separation is not completely obtained in one extraction step, complete extraction may be achieved by separated extractions.

- 1 Take a series of tubes (0, 1, 2, 3) containing a fixed amount of an aqueous solution of known pH.
- 2 Place the mixture in aq. solution to be separated in tube 0 and add fixed amount of Chloroform.
- 3 Shake the tube 0 to achieve distribution and allow the layers to separate.
- 4 Transfer the chloroform layer to tube 1 + add a fresh aliquot of Chloroform to tube 0.
- 5 Continue the cycle until separation is complete.

(v)

Purpose of Internal Standard

If the composition of sample and matrix is unknown. The internal standard is added to both unknown and calibration standards.

A standard different from analyte is added in constant amount to all samples, blanks and standards.

Useful when quantity of samples varies from run to run

Calibration plot = $\frac{S_x}{S_{is}}$ vs. C_{stds}

e.g. Li is used as internal standard for determination of Na or K in blood serum by ion chromatography.

(vi)

Distribution Ratio

1. The ratio of conc. of all the species of solute distributed b/w two immiscible solvents in contact with each other
2. It is represented by **D**
3. Example

$$D = \frac{[HBz]_e}{[HBz]_a + [Bz^-]_a}$$

4. This is the distribution ratio in which HBz is

Distribution co-efficient

1. The ratio of conc. of a solute in each of the solvent is always constant at a certain Temperature.
2. It is represented by **K** or **K_d**
3. Example

$$K_d = \frac{[HBz]_e}{[HBz]_a}$$

4. This the distribution co-efficient of Benzoic

Benzic acid, 'e' is ether layer and 'a' is aqueous layer. acid in ether + aqueous layer.

(vii)

Spectral Interference

It is caused by the radiation overlapping of absorption line due to emissions from another element or compound. It is also affected by the radiation of an emission line of another background radiation from flame, solvent or sample.

Many instruments can make measurements simultaneously on several lines for more than 70 elements in the same time it used to take the measurement on a single line/element combination.

ICP-OES have the ability to avoid spectral interference.

(viii) Gel Filtration

1. A type of size exclusion chromatography in which we use an aqueous solution as the mobile phase

2. Aq. solution such as Buffers

3. Stationary phase is composed of porous matrix of sephadex and agarose are useful as porous material.

4. Applicable for proteins and water soluble material.

Gel permeation Chromatography

It is a form of size exclusion chromatography in which we use as organic solvent as the mobile phase.

Organic solvents such as hexane and Toulene.

The Stationary phase is a porous matrix made up of compounds like cross-linked polystyrene, polyacrylamide gels, agarose gel etc.

Short Questions

(i)

Doppler Effect + Pressure broadening

A limitation to resolution in absorption spectroscopy is the Doppler effect, the broadening of the spectral lines in gases and vapours because of the motion of the atoms or molecules. The spectral line widths of absorption lines in gases should be very narrow. But the Doppler effect, caused by random thermal motion of the molecules, can broaden the observed linewidth considerably.

The doppler linewidth, which is proportional to the square root of the Temperature, can even be larger than the separation between neighboring narrow spectral lines. Thus, Doppler broadening can hide much of the fine structure contained in atomic or molecular spectra.

The shape of spectral lines due to collisions, referred to as pressure broadening. The collisions of other particles with the emitting light particle

interrupts the emission process and by shortening the characteristic time for the process, increases the uncertainty in energy emitted.

(ii)

Characteristic + Adv of weak anion resin.

Resins of weak base anion using groups like $-N^+H(R)_2OH^-$, $-N^+H_2(R)OH$, $-N^+H_3OH^-$.

These resins have weakly alkaline groups in the basic solution it does not dissociate & therefore has no exchange ability.

The advantage of weakly base resin is that it is easy to regenerate.

(iii)

Capillary Electrophoresis System

CE is an analytical technique used to separate ions on the basis of their electrophoretic mobility with the use of an applied voltage.

In this technique sample is injected in tube by following ways

Hydrodynamic injection
Electrokinetic injection.

(iv)

Oxidation State & Masking agent

The selectivity of extraction may be increased by modifying the oxidation state of the metal or impure. For example extraction of Iron from Chloride solution is prevented by reducing Fe(III) to Fe(II) , which does not extract.

Masking agents are also known as Metal complexing agent which prevent the particular metal from taking part in their usual reaction and these agent remove the interference of these metals without requiring actual separation.

The most common masking agents are Cyanide, Tartrate, Citrate, Fluoride and EDTA.

(v)

Percent Extraction.

The extraction efficiency is the percentage of solute that moves from one phase to the other. It is not more than 100%.

(vi)

Principle of AAS

For determination of particular metal in sample.

Absorption of light by elements to measure their concentration. It quantifies the absorption of ground state atoms into gaseous atoms.

Amount of absorption determines analyte concentration.

(vii)

Significance of Gel Chromatography

It can be used to separate compounds

such as small molecules, proteins, polysaccharides, nucleic acids.

Especially used in separation of enzymes.

In separation of Hydrocarbons.

Separation of Triglycerides.

Also estimates the molecular weight of large molecules.

Q 8 & 9 repeat

(x)

Solid Phase Extraction

In this technique hydrophobic organic functional groups of compound are chemically bonded to the solid surface. For example, the bonding of C_{18} chain on silica with particle size on the order of $40\mu m$.

The Vander Waal forces of attraction are developed b/w hydrophobic compound & solid surface.

(V)

Nebulization

Process of conversion of sample to a fine mist of finely dividing droplets using a jet of compound of compressed gas.

The solvent is first evaporated leaving finely divided solid particles. These solid particles move toward the flame, where the gaseous atoms and ions are produced.

The ions absorb the energy from flame & excited to high energy level. When the atoms return to ground state radiations of characteristic element are emitted.

(vii)
Air acetylene & Nitrous oxide

Air acetylene flame is stable, simple to operate and produce sufficient atomization to enable good sensitivity and freedom from interference from many elements.

The temp range of flame is $2150-2300^{\circ}\text{C}$ which is sufficient to break down the molecules of most elements to ground state atoms during the time the sample resides in the flame.

Nitrous oxide flame is hot and has no explosive hazard at Temp range is 3000°C . At such Temp. molecular dissociation of compound of such elements take place easily.

Q 2, 3, 4, 6, 9 repeat

Q 10 in Analytical group

(vii)

Chemical Interference

occur when an analyte is not totally decomposed in the flame. There are less atoms present and therefore reduced absorbance of the analyte cause interference.

Some interference appears to be vapour-phase effects and are presumably due to ~~occlusion~~ occlusion of analyte elements into particles of matrix. The use of platforms in furnaces, the use of reactive purge gases and dispersion of matrix can in some cases reduce interference.

(iv)

Softening of Hard H_2O

Hard water passes through a softener, the Ca^{2+} , Mg^{2+} replace with sodium ions. These ions held loosely and are replaced easily, free Na^+ also releases in H_2O .

Ion exchange resin starts with sodium ions stuck to it. As Hard H_2O passes through the column, Na^+ ions come off the resin and go into H_2O . while Ca^{2+} and Mg^{2+} come out and stick to resin.

(i)

Isoelectric Point

It is also known as Isoionic point.

"A point at which net charge on that specie is zero is known as Isoelectric point of particular species."

It is related to the pH.

It is represented by $pH(I)$, IEP, pI

Calculation

For example to calculate the isoelectric point of proteins. we calculate it from the pK_a value of protein.

(ii)

FP

Amount of light emitted by excited atom is measured.

Beer's Law is not obeyed.

Absorption intensity & signal response greatly influenced by Temp.

Atomization and excitation flame used.

Intensity vs Concentration data is obtained

Limited to Group I & II metals

AAS

Amount of light absorbed by ground state atom is measured.

Beer's law is obeyed.

Absorption intensity & signal response does not depend on Temp

Only Atomization flame is used.

Absorbance vs Concentration data is obtained

Useful for more than 70 metals